

Home assignment: day 9 → day 12

Deadline: 2026-07-14, 16:59. You can use sympy or any other software for symbolic computations.

1. I throw a fair coin until the sequence $HTHT$ appears.

- Draw the directed graph for this game.
- Setup a system of equations for the sequences with given endpoint and starting point at the start of the game.
- Setup a system of equations for the sequences with given starting point and endpoint at the end of the game.

Let N be the number of tosses, N_H — the number of H and N_T — the number of T till the end of the game.

- Find expected values $\mathbb{E}(N)$, $\mathbb{E}(N_H)$, $\mathbb{E}(N_T)$.
 - Find probabilities $\mathbb{P}(N = 12)$, $\mathbb{P}(N_H = 13)$, $\mathbb{P}(N_T = 14)$.
2. Alice and Bob throw a fair coin until the sequence $A = HHH$ or the sequence $B = THH$ appears. Alice wins if the sequence A appears first, Bob wins if the sequence B appears first.

- Draw the directed graph for this game.
- Setup a system of equations for the sequences with the given endpoint and starting point at the start of the game.
- Setup a system of equations for the sequences with the given starting point and endpoint at the end of the game.

Let N be the number of tosses, N_H — the number of H and N_T — the number of T till the end of the game.

- Find expected values $\mathbb{E}(N)$, $\mathbb{E}(N_H)$, $\mathbb{E}(N_T)$.
- What is the probability that Alice wins?
- Find probabilities $\mathbb{P}(N = 12)$, $\mathbb{P}(N_H = 13)$, $\mathbb{P}(N_T = 14)$.

Demo for the quiz on day 10

- I throw a fair coin until the sequence $HTHT$ appears. Draw the directed graph for this game.
 - Alice and Bob throw a fair coin until the sequence $A = HHH$ or the sequence $B = THH$ appears. Alice wins if the sequence A appears first, Bob wins if the sequence B appears first. Draw the directed graph for this game.
-